Answer Key - Biology - Undergraduate Level Assessment 24 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) The organism is extinct in the wild and cannot be harvested for vaccine production; B) The organism that causes smallpox is too costly to use in vaccine production; C) The use of the actual virulent variola virus may be dangerous if too much is administered; D) Smallpox virus is confined to research laboratories; E) There is no specific treatment for smallpox
   Note: The correct answer highlights that administering the virulent variola virus, which causes smallpox, poses significant health risks and dangers due to its pathogenicity, making it unsuitable for use in vaccination.

2. **Answer:** B
   Options: A) Commensalism; B) Succession; C) Mutualism; D) Parasitism
   Note: Succession is a process that describes the gradual change in species composition of an ecosystem over time, rather than an interaction between different species, which is what interspecies interactions entail.

3. **Answer:** A
   Options: A) The body controls water excretion through sweat glands.; B) Water excretion is determined by the concentration of electrolytes in the body's cells.; C) The kidneys control water excretion by changing the size of the glomerulus.; D) Water excreted is controlled by the hormone insulin.; E) The body adjusts water excretion by altering the pH level of the blood.
   Note: The body controls water excretion through sweat glands by regulating perspiration as a means of thermoregulation and fluid balance, allowing for the release of excess water in response to environmental conditions and physical activity.

4. **Answer:** D
   Options: A) Consistent response to antibiotics across all protozoan forms; B) Inability to adapt to environmental conditions; C) Absence of common ancestral form; D) Extreme diversity, taxonomic organization, possession of certain plant-like features; E) Indistinct cellular structures among different protozoan groups
   Note: Referring to protozoans as belonging to a single animal phylum overlooks their extreme diversity in forms and functions, complicates taxonomic organization due to their varied evolutionary relationships, and fails to recognize that some exhibit plant-like features, highlighting the inadequacy of a singular classification.

5. **Answer:** B
   Options: A) Cyanobacteria reduce sulfur compounds.; B) Cyanobacteria have no nuclei.; C) Green algae cannot reduce nitrogen compounds.; D) Green algae cannot photosynthesize.; E) Green algae produce cell membranes.

Note: Cyanobacteria are prokaryotic organisms that lack a defined nucleus, while green unicellular algae are eukaryotic and possess membrane-bound nuclei.

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because the correct sequence of chloroplast membranes, starting from the innermost layer where the thylakoids are located, followed by the inner membrane and finally the outer membrane, accurately reflects the structural organization of chloroplasts.

7. **Answer:** False
   Options: A) True; B) False
   Note: Proteins destined for the lysosome typically require multiple signals, including a signal peptide and a specific post-translational modification, such as mannose-6-phosphate tagging, for proper targeting and recognition by lysosomal receptors.

8. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because a frequency of 0.59 indicates that 59% of students are tasters, which is not a majority since a majority requires more than 50%.

9. **Answer:** False
   Options: A) True; B) False
   Note: Fungi lack chlorophyll and the necessary cellular structures to perform photosynthesis, relying instead on absorbing nutrients from organic matter.

10. **Answer:** False
    Options: A) True; B) False
    Note: The statement is false because while bacterial decomposition does release CO 2, it primarily contributes to the carbon cycle by breaking down organic matter into simpler compounds, which can then be utilized by other organisms, rather than directly increasing atmospheric carbon levels.

Long Form Response

11. **Answer:** Newborns can contract syphilis primarily through vertical transmission, where the bacterium Treponema pallidum is transmitted from an infected mother to her fetus during pregnancy. This can occur at any stage of pregnancy, but the risk is particularly high during the first trimester. Additionally, syphilis can be transmitted during childbirth if the baby comes into contact with infectious lesions or secretions. While the mention of contracting syphilis from a contaminated surface is not a typical transmission route for this disease, it is vital to understand that maternal health directly influences the risk of transmission; untreated syphilis in pregnant women can

lead to severe complications for both the mother and the baby, including stillbirth or congenital syphilis. Thus, early screening and treatment of syphilis in pregnant women are crucial for preventing transmission and safeguarding maternal and fetal health.
Note: The answer correctly identifies vertical transmission as the primary route for a newborn to contract syphilis from an infected mother during pregnancy and emphasizes the heightened risk during the first trimester, while also acknowledging the potential for transmission during childbirth through contact with infectious lesions.

12. **Answer:** Severing the dorsal root of a nerve in a rat's leg primarily affects sensory perception, leading to a complete loss of sensation in the affected area. This includes the inability to feel pain, temperature, and touch, as well as a disruption of reflex actions that rely on sensory input. Conversely, if the ventral root is severed, the rat retains sensory perception; it can still sense stimuli applied to the leg, but it loses the ability to respond with motor actions. This occurs because the ventral root is responsible for transmitting motor signals from the spinal cord to the muscles. Therefore, while sensory input remains intact with ventral root damage, the lack of motor output results in an inability to react, highlighting the distinct roles of the dorsal and ventral roots in the nervous system.
Note: correct because severing the dorsal root disrupts sensory pathways, resulting in loss of sensation and reflexes, while severing the ventral root affects motor pathways, leading to loss of motor response while preserving sensory perception.

End of Answer Key